

MY LEETCODE

REVISION HANDBOOK

Patterns • Complexity • Solutions • Recall

31 problems

Built for active recall before interviews — understand the pattern, then rebuild the code.

SMALL STEPS. REPEATED RECALL. STRONGER DSA.

Revision Index

Find a problem by number, pattern, or difficulty.

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PROBLEM 0001

#0001



1. Two Sum

Difficulty

Easy

Pattern

Hashing / Set

Language

C++

Chapter

Hashing / Set

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1. Two Sum

Easy

Hashing / Set

C++

QUESTION



Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`.



Example

Input: `nums = [2,7,11,15]`, `target = 9`
Output: `[0,1]` Explanation: Because `nums[0] + nums[1] == 9`, we return `[0, 1]`.



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ avg.

SPACE COMPLEXITY

 $O(n)$ 

PATTERN

Hashing / Set

</> Solution (C++)

Problem 0001

```
class Solution {
public:
    vector<int> twoSum(vector<int>& nums, int target) {
        unordered_map<int, int> mp;
        for (int i = 0; i < nums.size(); i++) {
            int x = target - nums[i];
            if (mp.find(x) != mp.end()) {
                return {mp[x], i};
            }
            mp[nums[i]] = i;
        }
        return {};
    }
};
```

#0002



2. Add Two Numbers

Difficulty

Medium

Pattern

Linked List

Language

C++

Chapter

Linked List

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PROBLEM 0002

2. Add Two Numbers

Medium

Linked List

C++

QUESTION



You are given two non-empty linked lists representing two non-negative integers. The digits are stored in reverse order, and each of their nodes contains a single digit. Add the two numbers and return the sum as a linked list.



Example

Input: l1 = [2,4,3], l2 = [5,6,4] Output: [7,0,8] Explanation: 342 + 465 = 807.



Visual Representation



pattern → nodes → result



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Linked List

</> Solution (C++)

Problem 0002

```

/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode() : val(0), next(nullptr) {}
 *     ListNode(int x) : val(x), next(nullptr) {}
 *     ListNode(int x, ListNode *next) : val(x), next(next) {}
 * };
 */
class Solution {
public:
    ListNode* addTwoNumbers(ListNode* l1, ListNode* l2) {
        ListNode* dummy = new ListNode(0);
        ListNode* temp = dummy;
        int carry = 0;
        while (l1 != NULL || l2 != NULL || carry != 0) {
            int sum = carry;
            if (l1 != NULL) {
                sum += l1->val;
                l1 = l1->next;
            }
            if (l2 != NULL) {
                sum += l2->val;
                l2 = l2->next;
            }
            carry = sum / 10;
            temp->next = new ListNode(sum % 10);
            temp = temp->next;
        }
        return dummy->next;
    }
};

```

#0003



3. Longest Substring Without Repeating Characters

Difficulty

Medium

Pattern

Hashing / Set

Language

C++

Chapter

Hashing / Set

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PROBLEM 0003

3. Longest Substring Without Repeating Characters

Medium

Hashing / Set

C++

QUESTION



Given a string s , find the length of the longest substring without duplicate characters.



Example

Input: $s = \text{"abcabcbb"}$ Output: 3
 Explanation: The answer is "abc", with the length of 3. Note that "bca" and "cab" are also correct answers.



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ avg.

SPACE COMPLEXITY

 $O(n)$ 

PATTERN

Hashing / Set

</> Solution (C++)

Problem 0003

```
class Solution {
public:
    int lengthOfLongestSubstring(string s) {
        unordered_set<char> st;
        int left = 0;
        int maxi = 0;
        for (int right = 0; right < s.size(); right++) {
            while (st.find(s[right]) != st.end()) {
                st.erase(s[left]);
                left++;
            }
            st.insert(s[right]);
            maxi = max(maxi, right - left + 1);
        }
        return maxi;
    }
};
```

PROBLEM 0004

#0004



4. Median of Two Sorted Arrays

Difficulty

Hard

Pattern

Sorting

Language

C++

Chapter

Sorting

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4. Median of Two Sorted Arrays

Hard

Sorting

C++

QUESTION



Given two sorted arrays `nums1` and `nums2` of size `m` and `n` respectively, return the median of the two sorted arrays.



Example

Input: `nums1 = [1,3]`, `nums2 = [2]` Output: 2.00000
 Explanation: merged array = `[1,2,3]` and median is 2.



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n \log n)$ 

SPACE COMPLEXITY

 $O(n) / O(1)$ 

PATTERN

Sorting

</> Solution (C++)

Problem 0004

```
class Solution {
public:
    double findMedianSortedArrays(vector<int>& nums1, vector<int>& nums2) {
        nums1.insert(nums1.end(), nums2.begin(), nums2.end());
        sort(nums1.begin(), nums1.end());
        if(nums1.size()%2==0){
            return (nums1[(nums1.size()/2)-1]+nums1[nums1.size()/2])/2.0;
        }
        else{
            return nums1[nums1.size()/2.0];
        }
        return{};
    }
};
```

PROBLEM 0007

#0007



7. Reverse Integer

Difficulty

Medium

Pattern

Math / Simulation

Language

C++

Chapter

Math / Simulation

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7. Reverse Integer

Medium

Math / Simulation

C++

QUESTION



Given a signed 32-bit integer x , return x with its digits reversed. If reversing x causes the value to go outside the signed 32-bit integer range $[-2^{31}, 2^{31} - 1]$, then return 0.



Example

Input: $x = 123$ Output: 321

Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Math / Simulation

</> Solution (C++)

Problem 0007

```
class Solution {
public:
    int reverse(int x) {
        int rev = 0;

        while(x!=0){
            int d = x%10;
            if(rev > INT_MAX/10 || rev < INT_MIN/10)
                return 0;
            rev = rev*10 + d;
            x=x/10;
        }
        return rev;
    }
};
```

PROBLEM 0009

#0009



9. Palindrome Number

Difficulty

Easy

Pattern

Math / Simulation

Language

C++

Chapter

Math / Simulation

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9. Palindrome Number

Easy

Math / Simulation

C++

QUESTION

- ✓ Given an integer x , return true if x is a palindrome, and false otherwise.

Example

Input: $x = 121$ Output: true Explanation: 121 reads as 121 from left to right and from right to left.

Visual Representation

INPUT

PATTERN

OUTPUT

TIME COMPLEXITY
 $O(n)$ SPACE COMPLEXITY
 $O(1)$ PATTERN
Math / Simulation

</> Solution (C++)

Problem 0009

```
class Solution {
public:
    bool isPalindrome(int x){
        if (x < 0)
            return false;
        if (x != 0 && x % 10 == 0)
            return false;
        int rev = 0;
        while (x > rev) {
            rev = rev * 10 + x % 10;
            x /= 10;
        }
        if(x==rev || x==rev/10)
            return true;
        else
            return false;
    }
};
```

#0011

**11. Container With Most Water****Difficulty**

Medium

Pattern

Two Pointers

Language

C++

Chapter

Two Pointers

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PROBLEM 0011

11. Container With Most Water

Medium

Two Pointers

C++

QUESTION

You are given an integer array `height` of length `n`. There are `n` vertical lines drawn such that the two endpoints of the `i`th line are `(i, 0)` and `(i, height[i])`.

**Example**

Input: `height = [1,8,6,2,5,4,8,3,7]` Output: 49
 Explanation: The above vertical lines are represented by array `[1,8,6,2,5,4,8,3,7]`. In this case, the max area of water (blue section) the container can contain is 49.

**Visual Representation**

INPUT

PATTERN

OUTPUT

**TIME COMPLEXITY****O(n)****SPACE COMPLEXITY****O(1)****PATTERN****Two Pointers****</> Solution (C++)**

Problem 0011

```
class Solution {
public:
    int maxArea(vector<int>& height) {
        int left = 0;
        int right = height.size() - 1;
        int area = 0;
        while (left < right) {
            int h = min(height[left], height[right]);
            int a = h * (right - left);
            area = max(a, area);
            if (height[left] < height[right]) {
                left++;
            } else {
                right--;
            }
        }
        return area;
    }
};
```

#0014



14. Longest Common Prefix

Difficulty

Easy

Pattern

Arrays / Implement...

Language

C++

Chapter

Arrays / Implement...

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PROBLEM 0014

14. Longest Common Prefix

Easy

Arrays / Implementation

C++

QUESTION



Write a function to find the longest common prefix string amongst an array of strings.



Example

Input: strs = ["flower", "flow", "flight"]
Output: "fl"



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY
O(n)



SPACE COMPLEXITY
O(1)



PATTERN
Arrays / Implementation

</> Solution (C++)

Problem 0014

```
class Solution {
public:
    string longestCommonPrefix(vector<string>& strs) {

        string prefix = strs[0];

        for(int i = 1; i < strs.size(); i++) {

            int j = 0;

            while(j < prefix.size() &&
                  j < strs[i].size() &&
                  prefix[j] == strs[i][j]) {
                j++;
            }

            prefix = prefix.substr(0, j);

            if(prefix == "")
                return "";
        }

        return prefix;
    }
};
```

#0019



19. Remove Nth Node From End of List

Difficulty

Medium

Pattern

Linked List

Language

C++

Chapter

Linked List

QUICK LINKS

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PROBLEM 0019

19. Remove Nth Node From End of List

Medium

Linked List

C++

QUESTION



Given the head of a linked list, remove the n th node from the end of the list and return its head.



Example

Input: head = [1,2,3,4,5], n = 2 Output: [1,2,3,5]



Visual Representation



pattern → nodes → result



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Linked List

</> Solution (C++)

Problem 0019

```

/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode() : val(0), next(nullptr) {}
 *     ListNode(int x) : val(x), next(nullptr) {}
 *     ListNode(int x, ListNode *next) : val(x), next(next) {}
 * };
 */
class Solution {
public:
    ListNode* removeNthFromEnd(ListNode* head, int n) {
        ListNode *dummy = new ListNode(0);
        dummy->next = head;
        ListNode *temp=head;
        int count = 0;
        while(temp!=NULL){
            temp = temp->next;
            count++;
        }
        int k=count-n;
        temp = dummy;
        while(k>0){
            temp=temp->next;
            k--;
        }
        ListNode *temp1=temp->next;
        temp->next=temp1->next;
        delete temp1;
        ListNode* newHead = dummy->next;
        delete dummy;
        return newHead;
    }
};

```

PROBLEM 0021

#0021



21. Merge Two Sorted Lists

Difficulty

Easy

Pattern

Linked List

Language

C++

Chapter

Linked List

QUICK LINKS

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21. Merge Two Sorted Lists

Easy

Linked List

C++

QUESTION

- ✓ You are given the heads of two sorted linked lists list1 and list2.

Example

Input: list1 = [1,2,4], list2 = [1,3,4]
Output: [1,1,2,3,4,4]

Visual Representation



pattern → nodes → result



TIME COMPLEXITY
O(n)



SPACE COMPLEXITY
O(1)



PATTERN
Linked List

</> Solution (C++)

Problem 0021

```

/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode() : val(0), next(nullptr) {}
 *     ListNode(int x) : val(x), next(nullptr) {}
 *     ListNode(int x, ListNode *next) : val(x), next(next) {}
 * };
 */
class Solution {
public:
    ListNode* mergeTwoLists(ListNode* list1, ListNode* list2) {
        if(list1==NULL){return list2;}
        if(list2==NULL){return list1;}
        ListNode* dummy=new ListNode(-1);
        ListNode* temp = dummy;
        while(list1!=NULL && list2!=NULL){
            if(list1->val<=list2->val){
                temp->next=list1;
                list1=list1->next;
            }
            else{
                temp->next=list2;
                list2=list2->next;
            }
            temp=temp->next;
        }
        if(list1!=NULL){
            temp->next=list1;
        }
        else{
            temp->next=list2;
        }
        return dummy->next;
    }
};

```

PROBLEM 0024

#0024



24. Swap Nodes in Pairs

Difficulty

Medium

Pattern

Linked List

Language

C++

Chapter

Linked List

QUICK LINKS

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- > Notes
- > Similar Problems

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24. Swap Nodes in Pairs

Medium

Linked List

C++

QUESTION



Given a linked list, swap every two adjacent nodes and return its head. You must solve the problem without modifying the values in the list's nodes (i.e., only nodes themselves may be changed.)



Example

Input → Output



Visual Representation



pattern → nodes → result



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Linked List

</> Solution (C++)

Problem 0024

```

/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode() : val(0), next(nullptr) {}
 *     ListNode(int x) : val(x), next(nullptr) {}
 *     ListNode(int x, ListNode *next) : val(x), next(next) {}
 * };
 */
class Solution {
public:
    ListNode* swapPairs(ListNode* head) {
        ListNode* dummy = new ListNode(0);
        dummy->next = head;
        ListNode* temp = dummy;
        while(temp->next != NULL && temp->next->next != NULL) {
            ListNode* first = temp->next;
            ListNode* second = first->next;
            first->next = second->next;
            second->next = first;
            temp->next = second;
            temp = first;
        }
        ListNode* newHead = dummy->next;
        delete dummy;
        return newHead;
    }
};

```

PROBLEM 0033

#0033



33. Search in Rotated Sorted Array

Difficulty

Medium

Pattern

Arrays / Implement...

Language

C++

Chapter

Arrays / Implement...

QUICK LINKS

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33. Search in Rotated Sorted Array

Medium

Arrays / Implementation

C++

QUESTION

There is an integer array `nums` sorted in ascending order (with distinct values).

Example

Input: `nums = [4,5,6,7,0,1,2]`, `target = 0`
Output: `4`



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Arrays / Implementation

</> Solution (C++)

Problem 0033

```
class Solution {
public:
    int search(vector<int>& nums, int target) {
        for(int i=0;i<nums.size();i++){
            if(nums[i]==target){
                return i;
            }
        }
        return -1;
    }
};
```

#0034

**34. Find First and Last Position of Element in Sorted Array**

Difficulty

Medium

Pattern

Binary Search

Language

C++

Chapter

Binary Search

QUICK LINKS

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PROBLEM 0034

34. Find First and Last Position of Element in Sorted Array

Medium

Binary Search

C++

QUESTION



Given an array of integers `nums` sorted in non-decreasing order, find the starting and ending position of a given target value.

**Example**

Input: `nums = [5,7,7,8,8,10]`, `target = 8`
Output: `[3,4]`

**Visual Representation**

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Binary Search</> **Solution (C++)**

Problem 0034

```
class Solution {
public:
    vector<int> searchRange(vector<int>& nums, int target) {
        int first = -1;
        int last = -1;
        // Find first occurrence
        int left = 0;
        int right = nums.size() - 1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (nums[mid] == target) {
                first = mid;
                right = mid - 1; // search left
            }
            else if (target > nums[mid]) {
                left = mid + 1;
            }
            else {
                right = mid - 1;
            }
        }
        // Find last occurrence
        left = 0;
        right = nums.size() - 1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (nums[mid] == target) {
                last = mid;
                left = mid + 1; // search right
            }
            else if (target > nums[mid]) {
                left = mid + 1;
            }
            else {
                right = mid - 1;
            }
        }
        return {first, last};
    }
};
```

PROBLEM 0035

#0035



35. Search Insert Position

Difficulty

Easy

Pattern

Binary Search

Language

C++

Chapter

Binary Search

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35. Search Insert Position

Easy

Binary Search

C++

QUESTION



Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.



Example

Input: nums = [1,3,5,6], target = 5 Output: 2



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Binary Search

</> Solution (C++)

Problem 0035

```
class Solution {
public:
    int searchInsert(vector<int>& nums, int target) {
        int left = 0;
        int right = nums.size()-1;
        while(left<=right){
            int mid = left + (right - left)/2;
            if(nums[mid]==target){
                return mid;
            }
            if(target>nums[mid]){
                left=mid+1;
            }
            else{
                right=mid-1;
            }
        }
        return left;
    }
};
```

PROBLEM 0049

#0049



49. Group Anagrams

Difficulty

Medium

Pattern

Hashing / Set

Language

C++

Chapter

Hashing / Set

QUICK LINKS

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49. Group Anagrams

Medium

Hashing / Set

C++

QUESTION



Given an array of strings `strs`, group the anagrams together. You can return the answer in any order.



Example

Input → Output



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n \log n)$ 

SPACE COMPLEXITY

 $O(n) / O(1)$ 

PATTERN

Hashing / Set

</> Solution (C++)

Problem 0049

```
class Solution {
public:
    vector<vector<string>> groupAnagrams(vector<string>& strs) {
        unordered_map<string, vector<string>> mp;
        for(string s: strs){
            string key = s;
            sort(key.begin(), key.end());
            mp[key].push_back(s);
        }
        vector<vector<string>> ans;
        for(auto x: mp){
            ans.push_back(x.second);
        }
        return ans;
    }
};
```

PROBLEM 0053

#0053

**53. Maximum Subarray****Difficulty**

Medium

Pattern

Greedy / Kadane

Language

C++

Chapter

Greedy / Kadane

QUICK LINKS

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53. Maximum Subarray

Medium

Greedy / Kadane

C++

QUESTION

Given an integer array `nums`, find the subarray with the largest sum, and return its sum.

**Example**

Input: `nums = [-2,1,-3,4,-1,2,1,-5,4]`
 Output: 6 Explanation: The subarray `[4,-1,2,1]` has the largest sum 6.

**Visual Representation**

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Greedy / Kadane**</> Solution (C++)**

Problem 0053

```
class Solution {
public:
    int maxSubArray(vector<int>& nums) {
        int sum = 0;
        int maxsum = INT_MIN;
        for(int i = 0; i < nums.size(); i++){
            sum += nums[i];
            maxsum = max(maxsum, sum);
            if(sum < 0){
                sum = 0;
            }
        }
        return maxsum;
    }
};
```

#0076



76. Minimum Window Substring

Difficulty

Hard

Pattern

Hashing / Set

Language

C++

Chapter

Hashing / Set

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PROBLEM 0076

76. Minimum Window Substring

Hard

Hashing / Set

C++

QUESTION



Given two strings s and t of lengths m and n respectively, return the minimum window substring of s such that every character in t (including duplicates) is included in the window. If there is no such substring, return the empty string $""$.



Example

Input: $s = \text{"ADOBECODEBANC"}, t = \text{"ABC"}$
 Output: "BANC" Explanation: The minimum window substring "BANC" includes 'A', 'B', and 'C' from string t .



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ avg.

SPACE COMPLEXITY

 $O(n)$ 

PATTERN

Hashing / Set

</> Solution (C++)

Problem 0076

```
class Solution {
public:
    string minWindow(string s, string t) {
        unordered_map<char,int> n,w;
        for(char c:t) n[c]++;
        int l = 0,start=0;
        int req = n.size();
        int have = 0;
        int len = INT_MAX ;
        for(int r = 0; r<s.size(); r++){
            w[s[r]]++;
            if(n.count(s[r]) && w[s[r]]==n[s[r]]){
                have++;
            }
            while(have == req){
                if(r-l+1 < len){
                    len = r-l+1;
                    start = l;
                }
                w[s[l]]--;
                if(n.count(s[l])&&w[s[l]]<n[s[l]]){
                    have--;
                }
                l++;
            }
        }
        if(len == INT_MAX) return "";
        return s.substr(start,len);
    }
};
```

#0081



81. Search in Rotated Sorted Array II

Difficulty

Medium

Pattern

Binary Search

Language

C++

Chapter

Binary Search

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
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PROBLEM 0081

81. Search in Rotated Sorted Array II

Medium

Binary Search

C++

QUESTION



There is an integer array `nums` sorted in non-decreasing order (not necessarily with distinct values).



Example

Input: `nums = [2,5,6,0,0,1,2]`, `target = 0`
Output: `true`



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Binary Search

</> Solution (C++)

Problem 0081

```
class Solution {
public:
    bool search(vector<int>& nums, int target) {
        int left = 0;
        int right = nums.size() - 1;
        while (left <= right) {
            int mid = left + (right - left) / 2;
            if (nums[mid] == target)
                return true;
            // Duplicates: cannot determine which half is sorted
            if (nums[left] == nums[mid] && nums[mid] == nums[right]) {
                left++;
                right--;
            }
            // Left half is sorted
            else if (nums[left] <= nums[mid]) {
                if (nums[left] <= target && target < nums[mid])
                    right = mid - 1;
                else
                    left = mid + 1;
            }
            // Right half is sorted
            else {
                if (nums[mid] < target && target <= nums[right])
                    left = mid + 1;
                else
                    right = mid - 1;
            }
        }
        return false;
    }
};
```

PROBLEM 0094

#0094



94. Binary Tree Inorder Traversal

Difficulty

Easy

Pattern

Binary Search

Language

C++

Chapter

Binary Search

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
- > [Similar Problems](#)

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94. Binary Tree Inorder Traversal

Easy

Binary Search

C++

QUESTION

- Given the root of a binary tree, return the inorder traversal of its nodes' values.



Example

Input → Output



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

$O(n)$



SPACE COMPLEXITY

$O(1)$



PATTERN

Binary Search

</> Solution (C++)

Problem 0094

```
/**
 * Definition for a binary tree node.
 * struct TreeNode {
 *     int val;
 *     TreeNode *left;
 *     TreeNode *right;
 *     TreeNode() : val(0), left(nullptr), right(nullptr) {}
 *     TreeNode(int x) : val(x), left(nullptr), right(nullptr) {}
 *     TreeNode(int x, TreeNode *left, TreeNode *right) : val(x), left(left), right(right) {}
 * };
 */
class Solution {
public:
    vector<int> ans;
    vector<int> inorderTraversal(TreeNode* root) {
        if(root == NULL)
            return ans;
        inorderTraversal(root->left);
        ans.push_back(root->val);
        inorderTraversal(root->right);
        return ans;
    }
};
```

#0098



98. Validate Binary Search Tree

Difficulty

Medium

Pattern

Binary Search

Language

C++

Chapter

Binary Search

QUICK LINKS

- > [LeetCode](#)
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PROBLEM 0098

98. Validate Binary Search Tree

Medium

Binary Search

C++

QUESTION



Given the root of a binary tree, determine if it is a valid binary search tree (BST).



Example

Input: root = [2,1,3] Output: true



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Binary Search

</> Solution (C++)

Problem 0098

```
/**
 * Definition for a binary tree node.
 * struct TreeNode {
 *     int val;
 *     TreeNode *left;
 *     TreeNode *right;
 *     TreeNode() : val(0), left(nullptr), right(nullptr) {}
 *     TreeNode(int x) : val(x), left(nullptr), right(nullptr) {}
 *     TreeNode(int x, TreeNode *left, TreeNode *right) : val(x), left(left), right(right) {}
 * };
 */
class Solution {
public:
    bool check(TreeNode* root, long long low, long long high){
        if(root == NULL) return true;
        if (root->val <= low || root->val >= high)
            return false;
        return check(root->left, low, root->val)&&check(root->right, root->val, high);
    }
    bool isValidBST(TreeNode* root) {
        return check(root, LLONG_MIN, LLONG_MAX);
    }
};
```

PROBLEM 0100

#0100



100. Same Tree

Difficulty

Easy

Pattern

Binary Search

Language

C++

Chapter

Binary Search

QUICK LINKS

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- > Notes
- > Similar Problems

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100. Same Tree

Easy

Binary Search

C++

QUESTION



Given the roots of two binary trees p and q , write a function to check if they are the same or not.



Example

Input: $p = [1,2,3]$, $q = [1,2,3]$ Output: true



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Binary Search

</> Solution (C++)

Problem 0100

```
/**
 * Definition for a binary tree node.
 * struct TreeNode {
 *     int val;
 *     TreeNode *left;
 *     TreeNode *right;
 *     TreeNode() : val(0), left(nullptr), right(nullptr) {}
 *     TreeNode(int x) : val(x), left(nullptr), right(nullptr) {}
 *     TreeNode(int x, TreeNode *left, TreeNode *right) : val(x), left(left), right(right) {}
 * };
 */
class Solution {
public:
    bool isSameTree(TreeNode* p, TreeNode* q) {
        if(p==NULL && q==NULL) return true;
        if(p==NULL || q==NULL) return false;
        if(p->val!=q->val) return false;
        return isSameTree(p->left,q->left)&&isSameTree(p->right,q->right);
    }
};
```

PROBLEM 0101

#0101



101. Symmetric Tree

Difficulty

Easy

Pattern

Binary Search

Language

C++

Chapter

Binary Search

QUICK LINKS

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101. Symmetric Tree

Easy

Binary Search

C++

QUESTION



Given the root of a binary tree, check whether it is a mirror of itself (i.e., symmetric around its center).



Example

Input: root = [1,2,2,3,4,4,3] Output: true



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Binary Search

</> Solution (C++)

Problem 0101

```
/**
 * Definition for a binary tree node.
 * struct TreeNode {
 *     int val;
 *     TreeNode *left;
 *     TreeNode *right;
 *     TreeNode() : val(0), left(nullptr), right(nullptr) {}
 *     TreeNode(int x) : val(x), left(nullptr), right(nullptr) {}
 *     TreeNode(int x, TreeNode *left, TreeNode *right) : val(x), left(left), right(right) {}
 * };
 */
class Solution {
public:
    bool mirror(TreeNode* left, TreeNode* right) {
        if (left == NULL && right == NULL)
            return true;
        if (left == NULL || right == NULL)
            return false;
        if (left->val != right->val)
            return false;
        return mirror(left->left, right->right) &&
            mirror(left->right, right->left);
    }
    bool isSymmetric(TreeNode* root) {
        if (root == NULL)
            return true;
        return mirror(root->left, root->right);
    }
};
```

PROBLEM 0121

#0121

**121. Best Time to Buy and Sell Stock****Difficulty**

Easy

Pattern

Greedy / Kadane

Language

C++

Chapter

Greedy / Kadane

QUICK LINKS

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- > [Similar Problems](#)

Last Revised

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121. Best Time to Buy and Sell Stock

Easy

Greedy / Kadane

C++

QUESTION

You are given an array prices where prices[i] is the price of a given stock on the ith day.

**Example**

Input: prices = [7,1,5,3,6,4] Output: 5
 Explanation: Buy on day 2 (price = 1) and sell on day 5 (price = 6), profit = 6-1 = 5. Note that buying on day 2 and selling on day 1 is not allowed because you must buy before you sell.

**Visual Representation**

INPUT

PATTERN

OUTPUT

**TIME COMPLEXITY****O(n)****SPACE COMPLEXITY****O(1)****PATTERN****Greedy / Kadane****</> Solution (C++)**

Problem 0121

```
class Solution {
public:
    int maxProfit(vector<int>& prices) {
        int minprice = prices[0];
        int profit = 0;
        for(int i =1;i<prices.size();i++){
            minprice= min(minprice,prices[i]);
            profit = max(profit,prices[i]-minprice);
        }
        return profit;
    }
};
```

PROBLEM 0122

#0122

**122. Best Time to Buy and Sell Stock II****Difficulty**

Medium

Pattern

Greedy / Kadane

Language

C++

Chapter

Greedy / Kadane

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
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122. Best Time to Buy and Sell Stock II

Medium

Greedy / Kadane

C++

QUESTION

You are given an integer array `prices` where `prices[i]` is the price of a given stock on the `i`th day.

**Example**

Input: `prices = [7,1,5,3,6,4]` Output: 7
 Explanation: Buy on day 2 (price = 1) and sell on day 3 (price = 5), profit = 5-1 = 4. Then buy on day 4 (price = 3) and sell on day 5 (price = 6), profit = 6-3 = 3. Total profit is 4 + 3 = 7.

**Visual Representation**

INPUT

PATTERN

OUTPUT

**TIME COMPLEXITY****O(n)****SPACE COMPLEXITY****O(1)****PATTERN****Greedy / Kadane****</> Solution (C++)**

Problem 0122

```
class Solution {
public:
    int maxProfit(vector<int>& prices) {
        int profit=0;
        for(int i = 1;i<prices.size();i++){
            if(prices[i]>prices[i-1]){
                profit = profit + prices[i] - prices[i-1];
            }
        }
        return profit;
    }
};
```

#0128



128. Longest Consecutive Sequence

Difficulty

Medium

Pattern

Hashing / Set

Language

C++

Chapter

Hashing / Set

QUICK LINKS

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- > [Notes](#)
- > [Similar Problems](#)

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PROBLEM 0128

128. Longest Consecutive Sequence

Medium

Hashing / Set

C++

QUESTION



Given an unsorted array of integers `nums`, return the length of the longest consecutive elements sequence.



Example

Input: `nums = [100,4,200,1,3,2]` Output: 4
 Explanation: The longest consecutive elements sequence is `[1, 2, 3, 4]`. Therefore its length is 4.



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ avg.

SPACE COMPLEXITY

 $O(n)$ 

PATTERN

Hashing / Set

</> Solution (C++)

Problem 0128

```
class Solution {
public:
    int longestConsecutive(vector<int>& nums) {
        unordered_set<int> st(nums.begin(),nums.end());
        int ans =0;
        for(int x:st){
            if(st.find(x-1)==st.end()){
                int current =x;
                int count =1;
                while(st.find(current+1)!=st.end()){
                    current++;
                    count++;
                }
                ans=max(ans,count);
            }
        }
        return ans;
    }
};
```

PROBLEM 0189

#0189

**189. Rotate Array**

Difficulty

Medium

Pattern

Math / Simulation

Language

C++

Chapter

Math / Simulation

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
- > [Similar Problems](#)

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189. Rotate Array

Medium

Math / Simulation

C++

QUESTION



Given an integer array `nums`, rotate the array to the right by `k` steps, where `k` is non-negative.

**Example**

Input: `nums = [1,2,3,4,5,6,7]`, `k = 3`
Output: `[5,6,7,1,2,3,4]` Explanation: rotate 1 steps to the right: `[7,1,2,3,4,5,6]`
rotate 2 steps to the right: `[6,7,1,2,3,4,5]`
rotate 3 steps to the right: `[5,6,7,1,2,3,4]`

**Visual Representation**

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Math / Simulation</> **Solution (C++)**

Problem 0189

```
class Solution {
public:
    void rotate(vector<int>& nums, int k) {
        k = k % nums.size();
        reverse(nums.begin(), nums.end());
        reverse(nums.begin(), nums.begin() + k);
        reverse(nums.begin() + k, nums.end());
    }
};
```

#0209



209. Minimum Size Subarray Sum

Difficulty

Medium

Pattern

Arrays / Implement...

Language

C++

Chapter

Arrays / Implement...

QUICK LINKS

- > [LeetCode](#)
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PROBLEM 0209

209. Minimum Size Subarray Sum

Medium

Arrays / Implementation

C++

QUESTION

- ✓ Given an array of positive integers `nums` and a positive integer `target`, return the minimal length of a subarray whose sum is greater than or equal to `target`. If there is no such subarray, return 0 instead.

Example

Input: `target = 7, nums = [2,3,1,2,4,3]`
 Output: 2 Explanation: The subarray `[4,3]` has the minimal length under the problem constraint.

Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

O(n)

SPACE COMPLEXITY

O(1)

PATTERN

Arrays / Implementation

</> Solution (C++)

Problem 0209

```
class Solution {
public:
    int minSubArrayLen(int target, vector<int>& nums) {
        int l=0;
        int sum=0;
        int len = INT_MAX;
        for(int r=0;r<nums.size();++r){
            sum += nums[r];
            while(sum>=target){
                len = min(r-l+1,len);
                sum-=nums[l];
                ++l;
            }
        }
        if(len == INT_MAX) return 0;
        return len;
    }
};
```

#0215

**215. Kth Largest Element in an Array****Difficulty**

Medium

Pattern

Arrays / Implement...

Language

C++

Chapter

Arrays / Implement...

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
- > [Similar Problems](#)

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PROBLEM 0215

215. Kth Largest Element in an Array

Medium

Arrays / Implementation

C++

QUESTION

Given an integer array `nums` and an integer `k`, return the `k`th largest element in the array.

**Example**

Input: `nums = [3,2,1,5,6,4]`, `k = 2` Output: 5

**Visual Representation**

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ 

SPACE COMPLEXITY

 $O(1)$ 

PATTERN

Arrays / Implement...</> **Solution (C++)**

Problem 0215

```
class Solution {
public:
    int findKthLargest(vector<int>& nums, int k) {
        priority_queue<int, vector<int>, greater<int>> pq;
        for(int x:nums){
            pq.push(x);
            if(pq.size()>k){
                pq.pop();
            }
        }
        return pq.top();
    }
};
```

#0239



239. Sliding Window Maximum

Difficulty

Hard

Pattern

Two Pointers

Language

C++

Chapter

Two Pointers

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
- > [Similar Problems](#)

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PROBLEM 0239

239. Sliding Window Maximum

Hard

Two Pointers

C++

QUESTION

- ✓ You are given an array of integers `nums`, there is a sliding window of size `k` which is moving from the very left of the array to the very right. You can only see the `k` numbers in the window. Each time the sliding window moves right by one position.

Example

Input: `nums = [1,3,-1,-3,5,3,6,7]`, `k = 3`
 Output: `[3,3,5,5,6,7]` Explanation: Window position Max ----- [1 3 -1]
 -3 5 3 6 7 3 1 [3 -1 -3] 5 3 6 7 3 1 3 [-1 -3 5] 3 6 7 5 1 3 -1 [-3 5 3] 6 7 5 1 3 -1
 -3 [5 3 6] 7 6 1 3 -1 -3 5 [3 6 7] 7

Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY
 $O(n)$



SPACE COMPLEXITY
 $O(1)$



PATTERN
Two Pointers

</> Solution (C++)

Problem 0239

```
class Solution {
public:
    vector<int> maxSlidingWindow(vector<int>& nums, int k) {
        deque<int> dq;
        vector<int> ans;
        for (int r = 0; r < nums.size(); r++) {
            // Remove elements outside the window
            while (!dq.empty() && dq.front() <= r - k) {
                dq.pop_front();
            }
            // Remove smaller elements
            while (!dq.empty() && nums[dq.back()] <= nums[r]) {
                dq.pop_back();
            }
            // Add current index
            dq.push_back(r);
            // Window size is k
            if (r >= k - 1) {
                ans.push_back(nums[dq.front()]);
            }
        }
        return ans;
    }
};
```

PROBLEM 0455

#0455



455. Assign Cookies

Difficulty

Easy

Pattern

Sorting

Language

C++

Chapter

Sorting

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
- > [Similar Problems](#)

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455. Assign Cookies

Easy

Sorting

C++

QUESTION



Assume you are an awesome parent and want to give your children some cookies. But, you should give each child at most one cookie.



Example

Input: $g = [1, 2, 3]$, $s = [1, 1]$ Output: 1
 Explanation: You have 3 children and 2 cookies. The greed factors of 3 children are 1, 2, 3. And even though you have 2 cookies, since their size is both 1, you could only make the child whose greed



Visual Representation

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n \log n)$ 

SPACE COMPLEXITY

 $O(n) / O(1)$ 

PATTERN

Sorting

</> Solution (C++)

Problem 0455

```
class Solution {
public:
    int findContentChildren(vector<int>& g, vector<int>& s) {
        sort(g.begin(), g.end());
        sort(s.begin(), s.end());
        int i = 0;
        int j = 0;
        int count = 0;
        while (i < g.size() && j < s.size()) {
            if (s[j] >= g[i]) {
                count++;
                i++;
                j++;
            }
            else {
                j++;
            }
        }
        return count;
    }
};
```

PROBLEM 3718

#3718

**4080. Smallest Missing Multiple of K**

Difficulty

Easy

Pattern

Hashing / Set

Language

C++

Chapter

Hashing / Set

QUICK LINKS

- > [LeetCode](#)
- > [Notes](#)
- > [Similar Problems](#)

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4080. Smallest Missing Multiple of K

Easy

Hashing / Set

C++

QUESTION



Given an integer array `nums` and an integer `k`, return the smallest positive multiple of `k` that is missing from `nums`.

**Example**

Input → Output

**Visual Representation**

INPUT

PATTERN

OUTPUT



TIME COMPLEXITY

 $O(n)$ avg.

SPACE COMPLEXITY

 $O(n)$ 

PATTERN

Hashing / Set</> **Solution (C++)**

Problem 3718

```
class Solution {
public:
    int missingMultiple(vector<int>& nums, int k) {
        unordered_set<int> st;
        for(int x : nums){
            st.insert(x);
        }
        for(int i=1;i<=nums.size()+1;i++){
            int x = k*i;
            if(st.find(x)==st.end()){
                return x;
            }
        }
        return -1;
    }
};
```